

Visualizing the U.S. Treasury Yield Curve

An Interactive Data Visualization Dashboard

Abstract

My final product is a dashboard that is interactive and explores U.S. Treasury yield curve data from 2000 through 2024. The dashboard is written using Path C — it is fully API-driven and gets all the up-to-the-minute data from the U.S. Treasury Department's public XML endpoint, instead of loading from a local file each time. It was built with Python, Streamlit, and Plotly; and includes three main visualizations: the yield curve as a snapshot for any date selected, a historical time series chart for key maturities, and the spread analysis chart highlighting curve inversions.

1. Where the Data Comes From

The data is sourced from the U.S. Department of the Treasury public REST/XML API, which is called Daily Par Yield Curve Rates. This is a current end point, and will be updated daily on a business basis, reflecting the current yield rates on 1-month through 30-year Treasuries. API pipeline (Path C): Python's requests library is used to make HTTP GET requests directly to the Treasury API endpoint, as specified in the data module (data/fetch_data.py). XML response is cleaned and returned in the pandas DataFrame format. The result is stored in memory for 1 hour with the `@st.cache_data(ttl=3600)` decorator, so it will avoid hammering the API whenever widgets are interacted with and will also provide fresh data every time a session is run. No local database or static file — the data streams live from the Treasury's servers to the dashboard every time it is run. This source was selected because it is the one primary and authoritative source for the U.S. government's cost of borrowing, not a third party copy. There is no key or registration required for the API, it has 25 years of history and updates automatically each business day.

2. Why I Chose Each Visualization

2.1 Yield Curve Snapshot — Line Chart with Date Picker

The first chart depicts the yield curve on one day, with the maturity on the x axis and yield on the y axis. I also introduced a date picker, so that you can select any date from 2000 up to today and find out what the curve was like. I also added four preset buttons: Pre-GFC (2007), Post-GFC (2010), COVID low (2020), and Rate Hike Peak (2023). These allow you to switch between four entirely distinct economic modes and to compare them side by side.

I selected this chart because the shape of the curve for that single day reveals a lot about markets' expectations that day. If the slope is upwards and normal it indicates that the investors are confident. A flat or inverted curve indicates some problem. Each data point contains hover tooltips with exact yield at that maturity, meeting the rubric requirement of providing insight with hover.

The curve of the COVID era (mid-2020) is very different from all the other curves. The short end is almost flat at the low end as rates were reduced to zero by the Fed and the long end remained on the higher end. This chart readily displays that distinctive flat-then-rising graph.

2.2 Historical Yield Trends — Multi-Line Time Series

The second chart reveals the changes in yields for the 2-year, 10-year and 30-year Treasuries since 2000. I added a range slider at the bottom for zooming into each sub-period; and I shaded in red the NBER designated U.S. recessions for 2001, 2008-09, and 2020. Also, there's a toggle in the sidebar for showing or hiding which maturities to display — which is the toggle/select element that the rubric requires.

I selected this chart because I wanted to go with the question I had in mind: Has yield actually changed over 25 years? This secular drop from 2000 to 2020 is very evident in the time series; the 2022 increase is unusual. Below the chart, there is a summary statistics table (min, max, average, standard deviation) for quantitative context.

Finding: 10-year yield has experienced a 20-year downtrend from around 6.5% in 2000 to near 0.5% in 2020. The 5% turnaround in 2022 is the highest rate increase within the entire data set, growing from 1.5% to 5% in less than 2 years.

2.3 Spread Analysis — 10Y minus 2Y Area Chart

The third chart focuses on the 10-year minus 2-year yield spread. When this number goes negative, it means short-term borrowing costs exceed long-term ones — this is called a yield curve inversion. Economists consider this one of the most reliable recession warning signals: every U.S. recession since the 1970s has been preceded by at least one inversion.

I used green fill when the spread is positive and red fill when it is negative, so the signal is readable at a glance. A second line shows the 10Y-3M spread as a dotted overlay, which the Federal Reserve itself uses in its own research. Below the chart, a bar chart shows exactly how deep and long each inversion episode was.

Finding: The 2022-2023 inversion is the deepest in the dataset, reaching nearly -1.1 percentage points. That is worse than the pre-GFC inversion of 2006-07. Whether this will eventually lead to a recession, or whether the post-pandemic economy behaves differently, is still an open question.

3. Key Findings

- During this time of creating and playing around with this dashboard, I learned several things that I wasn't aware of before:
- The yield curve essentially reflects a sort of "snapshot of the collective wisdom of the bond markets. A ramp upwards = confidence. Flat or inverted = fear or uncertainty. It isn't a foolproof indicator, but there are other indicators and it's a strong one.
- This 2022 rate hike cycle was by far the most significant in the 25 years of data. What hasn't been experienced in this dataset before is the speed and the size of the move and it is very evident in each visualize; the time series, the heatmap, and the spread chart.
- Each of the three inversions in the data 2006-07, 2019, and 2022-23, were followed by some type of economic stress.
- This gradual decline in rates over the past two decades was not coincidental. It follows the macroeconomic trends, and includes declining inflation expectations, ageing population and the global need for safe assets like U.S. Treasuries. That trend has been partially reversed in the 2022 reversal.

4. If I Had Unlimited Resources — Future Directions

- I would extend the current live API architecture and go here with this project:
- Avoid directly calling Treasury APIs, instead use a data pipeline such as Tinybird or Supabase. The pipeline would consume the Treasury XML feed on a regular schedule, put it in a structured database and provide a clean RESTful service to the dashboard for it to query. This separates the front end from the source API, and provides reliability, versioning, and query speed.
- Implement a real-time websocket layer to keep the dashboard updated as Treasury data is added daily at night without page refresh.
- Use live macroeconomic data from the FRED API directly on the historic yield chart to better visualize monetary policy relationships, such as the CPI inflation rate, the unemployment rate, and Fed Funds rate.
- For a global perspective on rate cycles, include an international comparison tab that pulls yield curve data from the ECB, the Bank of England's and the Bank of Japan's APIs and displays their interactions with U.S. yields.
- Create a bond portfolio simulator that as the user enters his/her holdings, he or she will see how the duration risk and yield of a bond portfolio will evolve as the rate environment changes over time.

5. Collaboration and AI Usage

This project was done on my own. I primarily used Claude, an AI assistant from Anthropic, to help with debugging technical problems with Plotly and Streamlit.

In particular, it enabled me to find issues related to unsupported color formats and conflicting layout arguments in Plotly. I also leveraged it as a reference when configuring Streamlit styling and themes on the dashboard.

The main ideas for the project, the visualizations, the cost analysis, and the dashboard design were all created by me. AI support had to remain to a minimum—just technical debugging and UI support, not conceptual or analytical decisions.

Self-assessment: very good marks for the dashboard itself – the three core charts are all based on my original proposal, they all show some real insight and the design is clean and distinctive. One area where I found AI to be invaluable was in troubleshooting any syntax issues in the library, which I think is a fair use of the tool. All of the economic analysis and design options for visualization are my own.

References

1. U.S. Department of the Treasury. (2024). Daily Treasury Par Yield Curve Rates. Retrieved from <https://home.treasury.gov/resource-center/data-chart-center/interest-rates/>
2. Estrella, A., & Mishkin, F. S. (1998). Predicting U.S. recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1), 45-61.
3. Federal Reserve Bank of San Francisco. (2018). Does the yield curve really forecast recession? FRBSF Economic Letter 2018-20.
4. Mishkin, F. S. (1990). The information in the longer maturity term structure about future inflation. *Quarterly Journal of Economics*, 105(3), 815-828.
5. National Bureau of Economic Research. (2024). US Business Cycle Expansions and Contractions. Retrieved from <https://www.nber.org/research/business-cycle-dating>.